Lalit Maurya

+91-6364713301 | Bengaluru, India | lm742@snu.edu.in | linkedin.com/in/lalitm1004 | github.com/lalitm1004 | lm04.me

Profile

Computer Science undergraduate with experience in full-stack development and AI/ML, skilled in Rust, Python, and modern web technologies. Passionate about building scalable, real-world solutions that empower individuals and organizations.

Education

Shiv Nadar Institution of Eminence — Delhi NCR, India
Bachelor of Technology in Computer Science and Engineering (CGPA 8.39) — *Aug. 2023 – Expected May 2027*

Experience

Technology Consultant Intern — May 2025 – July 2025
Microsoft — *Hyderabad, India*

- Developed and deployed a production-ready 5-agent AI solution using Azure OpenAI, .NET 8/C#, Power Automate, and Dataverse, saving 5+ hrs/day per seller and reducing RFP turnaround from 2-3 days to under 1 minute; published to an internal enterprise AI hub for organization-wide reuse.
- Led development of an AI-powered sales enablement platform delivering real-time stakeholder insights, next-best-action recommendations, and automated executive-ready presentation decks from live enterprise data; built enterprise-grade services with layered architecture, Azure DevOps CI/CD, human-in-the-loop workflows, and cost-optimized inference.

Teaching Assistant - Computer Organisation and Architecture — Aug 2025 – Nov 2025
Shiv Nadar Institution of Eminence — *Delhi NCR, India*

Hackathon Director, @ping — April 2024 – Feb 2025
Google Developer Group — *Delhi NCR, India*

- Created and led @ping, a campus-wide hackathon initiative, organizing multiple hackathons across diverse formats and conducting bootcamps that helped students progress from beginner to intermediate development skills
- Led the development of reusable libraries and infrastructure for competitive programming events, including a Connect Four bot framework that enabled participants to build, test, and compete with their own AI agents
- Drove the initiative end-to-end, owning event ideation, technical direction, library design, bootcamp curriculum, and execution while leading the organizing team

Projects

oxigrad-autograd | *Rust, Python, PyO3, Maturin* — Jul 2025

- Developed a scalar-valued auto-differentiation engine in Rust for memory-safety and performance
- Implemented Python bindings using PyO3 and Maturin for seamless integration across languages
- Implemented forward and backward passes, supporting automatic differentiation and gradient computation through dynamic computation graphs
- Integrated common activation functions (Sigmoid, ReLU, Softmax) and loss functions (MSE, Cross Entropy)
- Published to PyPI with support for Python 3.9+, enabling simple installation via `pip` or `uv`

Skin Detection Without Color Information | *OpenCV, Scikit-learn, NumPy* — Feb 2025 - May 2025

- Implemented grayscale skin detection using facial priors and texture features from a WACV 2017 paper
- Processed and curated a HELEN-derived dataset with ground-truth skin segmentation masks
- Extracted LBP, grayscale stats, lacunarity, and landmark-based features
- Developed a region growing algorithm seeded by face-based intensity priors

QueenSweep | *Rust, Web Assembly, Typsecript, Chromium Extension* — Oct 2025 - Dec 2025

- Built a constraint satisfaction solver for LinkedIn Queens with aggressive pruning via 1-step lookahead and pluggable heuristics, solving 90% of puzzles in <10ms (p90: 9.55ms, 250+ states/ms)
- Designed zero-copy state representation using Rc<[T]> for immutable data sharing, reducing memory overhead by eliminating redundant clones during backtracking search
- Implemented WebAssembly bindings to compile Rust solver to browser-compatible binary, achieving near-native performance in JavaScript runtime
- Developed Chrome extension with DOM mutation observers for SPA navigation detection and synthetic event dispatch for automated solution application on queensgame.vercel.app

Technical Skills

Languages: Rust, JavaScript, TypeScript, Python, C, C++, SQL, HTML, CSS, C#
Frameworks & Libraries: React, Next.js, SvelteKit, Flask, FastAPI, PyTorch, Poem-OpenAPI, .NET 8/10
Tools & Platforms: Git, GitHub, Supabase, Docker, Google Cloud Platform (GCP), Azure, Azure Foundry, Linux, uv
Databases & ORMS: MySQL, PostgreSQL, MongoDB, Prisma, SQLx, Dataverse
Technologies & Protocols: Node.js, REST APIs, gRPC, WebSockets, Selenium